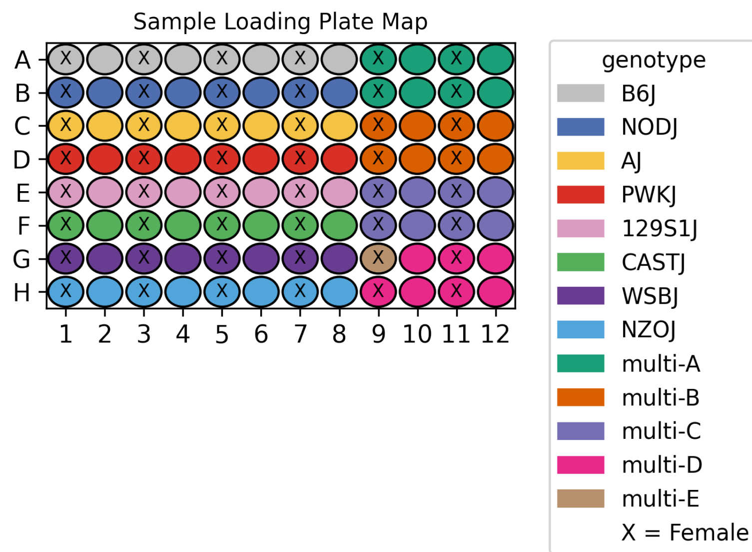
UCI/Caltech IGVF Dataset Pipeline Report

	igvf_011
Kit	WT_mega
Chemistry	Parse v2
Sample type	nuclei
Genotypes	B6J NODJ AJ PWKJ 129S1J CASTJ WSBJ NZOJ
Tissues	Gastrocnemius CortexHippocampus
Subpool	Subpool_13
Measurment Set	IGVFDS2892HFWJ
Exome capture subpool	No
Expected nuclei	67,000



1. Sample multiplexing

Gastrocnemius was the main tissue on the plate while CortexHippocampus was multiplexed.

Table 1: Genotype multiplexing key

Pair	Multiplexed Genotype 1	Multiplexed Genotype 2
multi-A	B6J	NODJ
multi-B	AJ	PWKJ
multi-C	129S1J	CASTJ
multi-D	NZOJ	WSBJ
multi-E	CASTJ	WSBJ

2. Alignment

A total of 1,557,744,959 reads were aligned using GENCODE vM32 with mm39 assembly.

Table 2: Pseudoalignment stats

Subpool	Total reads	Total UMIs	Fraction duplicated UMIs	Total pseudoaligned	Pct. pseudoaligned
Subpool_13	1,557,744,959	718,440,981	0.368	1,137,431,032	73.0

3. Cell recovery

A total of 187,099 nuclei in Subpool_13 have >=200 UMIs and 78,618 nuclei have >= 500 UMIs.

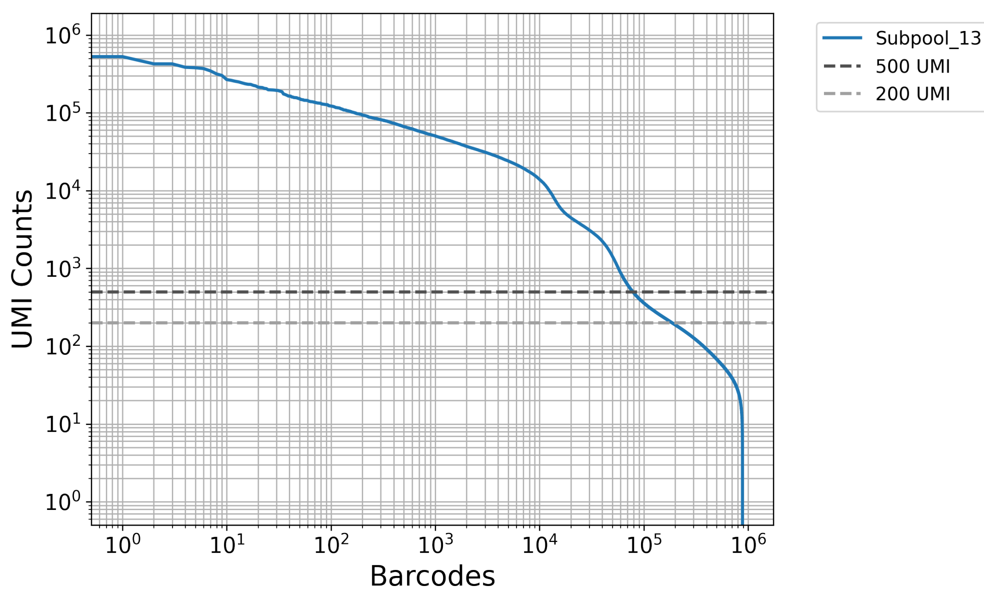


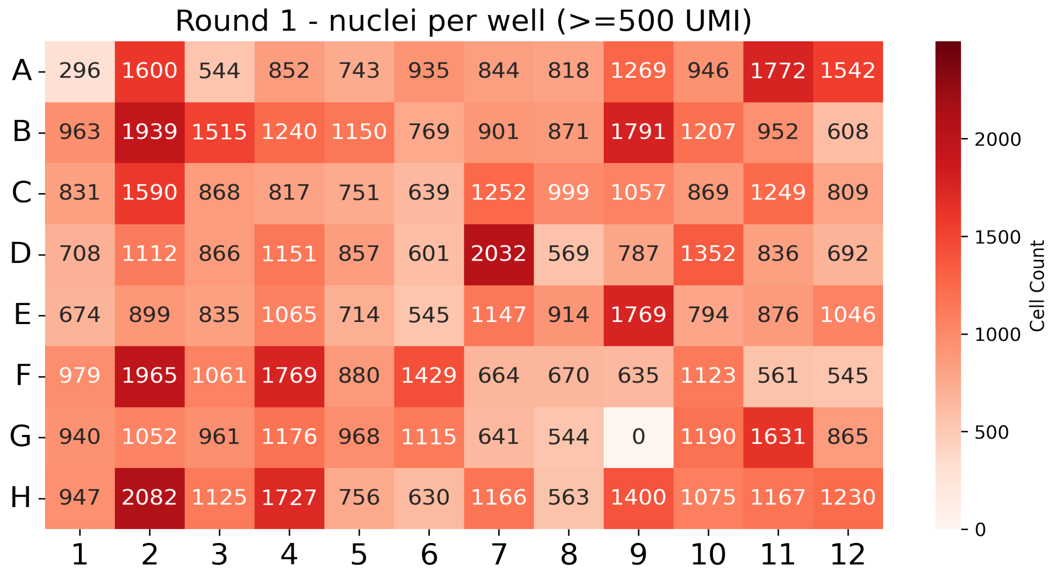
Table 3: QC stats for nuclei >=200 UMI

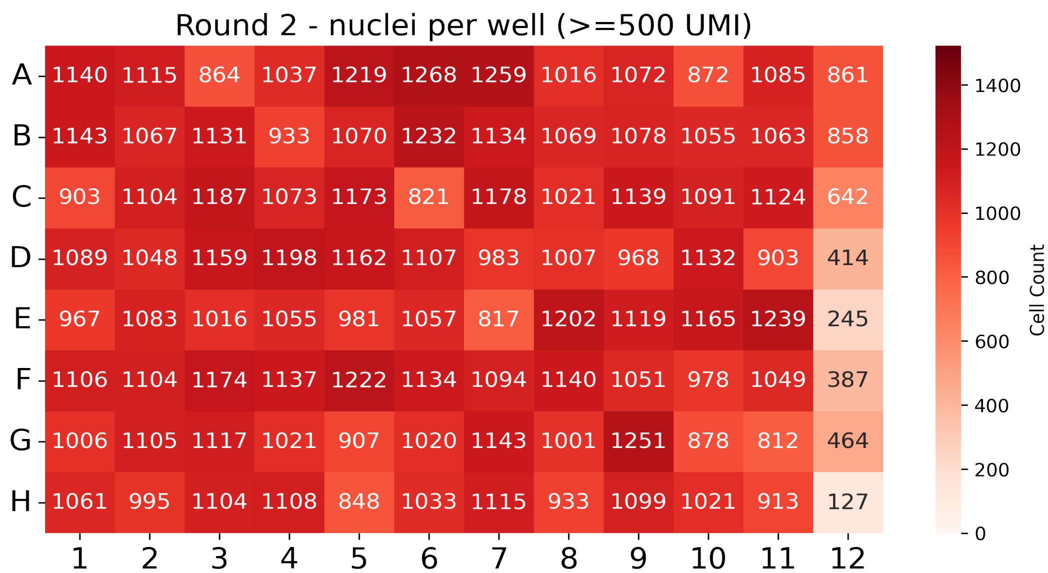
Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_13	187,099	390	2,777.0	276	831.8

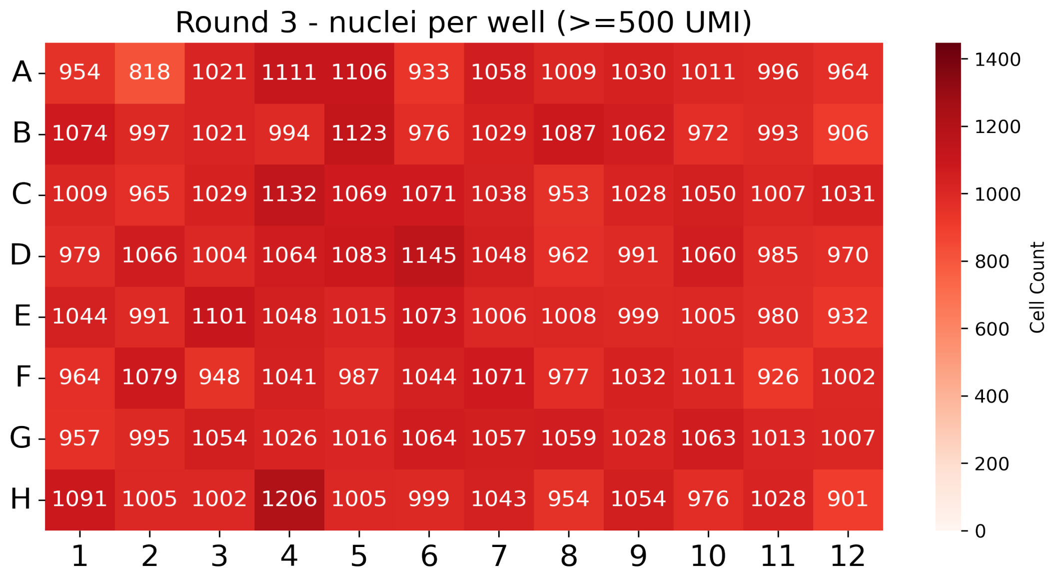
Table 4: QC stats for nuclei >=500 UMI

Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_13	78,618	2,296	6,206.8	1,111	1,677.9

In this combinatorial barcoding assay, nuclei are barcoded in 96-well plates over three rounds. The first round assigns sample-specific barcodes, while the next two are applied to the pooled mix. Even distribution in the first round is important to ensure consistent sample representation.







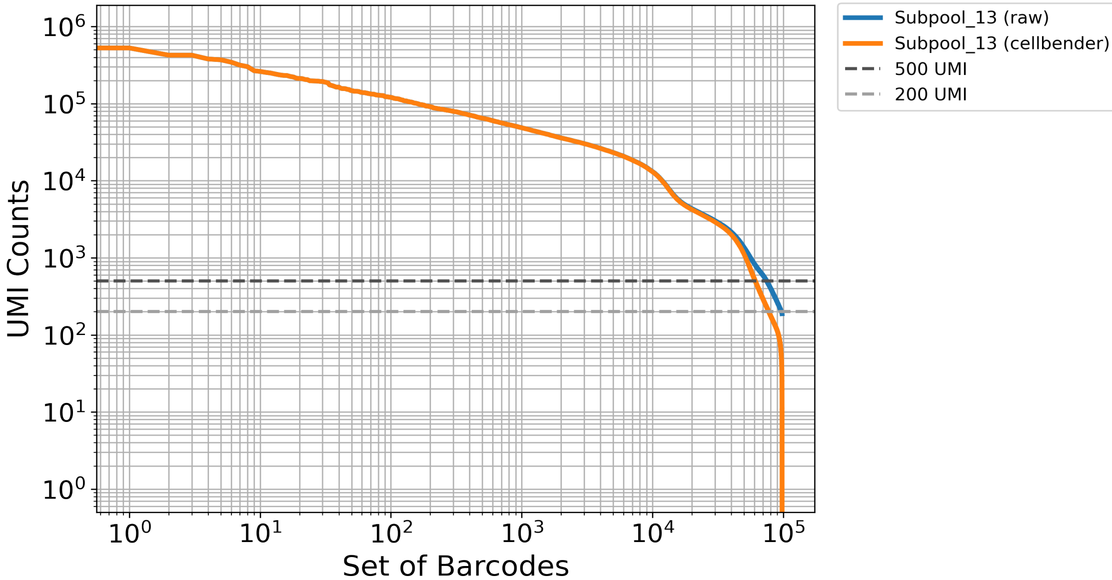
4. Background removal

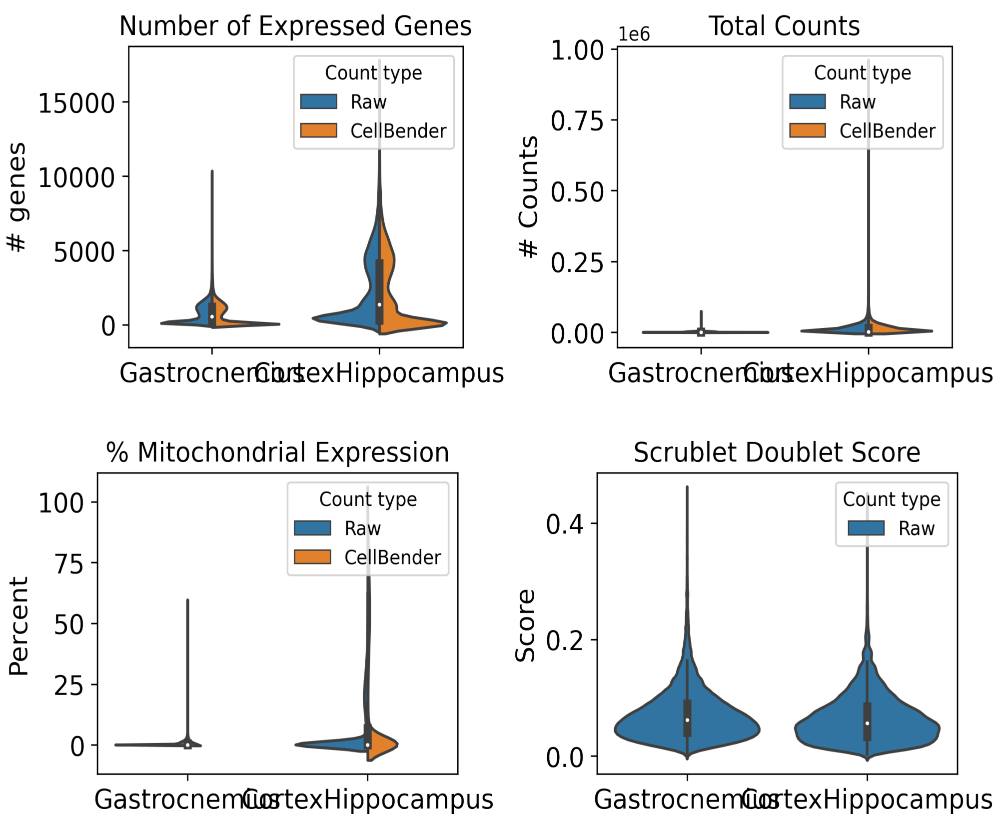
Table 5: Cellbender parameters

Subpool	droplets_included	learning_rate	expected_cells
Subpool_13	200000	5e-05	67000

Table 6: Cellbender stats

Subpool	Percent counts removed	Found nuclei	Mean denoised counts
Subpool_13	3.8	99,259	4,786.8





5. Barcode sample map

The first round corresponds to sample barcoding. In Parse Bio assays, two barcodes are included in the first round: a random hexamer (R) and oligo-dT (T) barcode per well (parse barcode type). This reduces 3-prime bias and allows for the capture of full-length transcripts. These sample level barcodes can be found in the barcode-1 region in the seqspec, and begin at position 16 (0-based) in the full 24-nt cell barcode. For the WT_mega kit, samples are distributed across 96 wells in round 1. The following table includes our lab sample ID as sample descriptions and IGVF sample accessions.

Table 7: Barcode sample map

barcode	sample accession	sample description	position	parse barcode type	well
CATCATCC	IGVFSM4575TTYT	016_B6J_10F_16	16	R	A1
CATTCCTA	IGVFSM4575TTYT	016_B6J_10F_16	16	T	A1
CTGCTTTG	IGVFSM2276CJGW	017_B6J_10M_16	16	R	A2
CTTCATCA	IGVFSM2276CJGW	017_B6J_10M_16	16	T	A2
CTAAGGGA	IGVFSM0465UALO	018_B6J_10F_16	16	R	A3
CCTATATC	IGVFSM0465UALO	018_B6J_10F_16	16	T	A3
GCTTATAG	IGVFSM3666GJLM	019_B6J_10M_16	16	R	A4
ACATTTAC	IGVFSM3666GJLM	019_B6J_10M_16	16	T	A4
TCTGATCC	IGVFSM4471UQDS	020_B6J_10F_16	16	R	A5
ACTTAGCT	IGVFSM4471UQDS	020_B6J_10F_16	16	T	A5
TCTCTTGG	IGVFSM0131WZLT	021_B6J_10M_16	16	R	A6
CCAATTCT	IGVFSM0131WZLT	021_B6J_10M_16	16	T	A6
CAATTTCC	IGVFSM5168ZHLV	024_B6J_10F_16	16	R	A7
GCCTATCT	IGVFSM5168ZHLV	024_B6J_10F_16	16	T	A7
AGTCTCTT	IGVFSM2944JZTR	025_B6J_10M_16	16	R	A8
ATGCTGCT	IGVFSM2944JZTR	025_B6J_10M_16	16	T	A8
TGCTGCTC	IGVFSM2747ZOJJ,IGVFSM9909OKAR	016_B6J_10F_03	16	R	A9
CATTTACA	IGVFSM2747ZOJJ,IGVFSM9909OKAR	016_B6J_10F_03	16	T	A9
TGCTGCTC	IGVFSM1810KYOB,IGVFSM5439QLWY	066_NODJ_10F_03	16	R	A9
CATTTACA	IGVFSM1810KYOB,IGVFSM5439QLWY	066_NODJ_10F_03	16	T	A9
GTATTTCC	IGVFSM5751LMFR,IGVFSM9616JAGA	017_B6J_10M_03	16	R	A10
ACTCGTAA	IGVFSM5751LMFR,IGVFSM9616JAGA	017_B6J_10M_03	16	T	A10
GTATTTCC	IGVFSM3477RPDP,IGVFSM6514BRQT	067_NODJ_10M_03	16	R	A10
ACTCGTAA	IGVFSM3477RPDP,IGVFSM6514BRQT	067_NODJ_10M_03	16	T	A10
TTCCTGTG	IGVFSM4584AIMA,IGVFSM6999WOUL	018_B6J_10F_03	16	R	A11
CCTTTGCA	IGVFSM4584AIMA,IGVFSM6999WOUL	018_B6J_10F_03	16	T	A11
TTCCTGTG	IGVFSM0483CXTW,IGVFSM2315DNPR	068_NODJ_10F_03	16	R	A11

CCTTTGCA	IGVFSM0483CXTW,IGVFSM2315DNPR	068_NODJ_10F_03	16	T	A11
GCTGCTTC	IGVFSM3502KEWR,IGVFSM9565QVTN	019_B6J_10M_03	16	R	A12
ACTCCTGC	IGVFSM3502KEWR,IGVFSM9565QVTN	019_B6J_10M_03	16	T	A12
GCTGCTTC	IGVFSM2210CUOC,IGVFSM3655BTER	069_NODJ_10M_03	16	R	A12
ACTCCTGC	IGVFSM2210CUOC,IGVFSM3655BTER	069_NODJ_10M_03	16	T	A12
TATGTGTC	IGVFSM4279QZLF	066_NODJ_10F_16	16	R	B1
ATTTGGCA	IGVFSM4279QZLF	066_NODJ_10F_16	16	T	B1
CAATTCTC	IGVFSM1252BKWG	067_NODJ_10M_16	16	R	B2
TTATTCTG	IGVFSM1252BKWG	067_NODJ_10M_16	16	T	B2
TGGTCTCC	IGVFSM8410WIND	074_NODJ_10F_16	16	R	B3
TCATGCTC	IGVFSM8410WIND	074_NODJ_10F_16	16	T	B3
GCTCTTTA	IGVFSM7620ICFV	069_NODJ_10M_16	16	R	B4
CATACTTC	IGVFSM7620ICFV	069_NODJ_10M_16	16	T	B4
GCTGCATG	IGVFSM3414NFLU	070_NODJ_10F_16	16	R	B5
CCGTTCTA	IGVFSM3414NFLU	070_NODJ_10F_16	16	T	B5
ACTCATTT	IGVFSM9301WUUD	071_NODJ_10M_16	16	R	B6
GCTTCATA	IGVFSM9301WUUD	071_NODJ_10M_16	16	T	B6
AGTCTTGG	IGVFSM5744JVUK	072_NODJ_10F_16	16	R	B7
CTCTGTGC	IGVFSM5744JVUK	072_NODJ_10F_16	16	T	B7
GGTTCTTC	IGVFSM6017WKVH	073_NODJ_10M_16	16	R	B8
CCCTTATA	IGVFSM6017WKVH	073_NODJ_10M_16	16	T	B8
TCATGTTG	IGVFSM6578SAUC,IGVFSM9764WOWB	020_B6J_10F_03	16	R	B9
ACTGCTCT	IGVFSM6578SAUC,IGVFSM9764WOWB	020_B6J_10F_03	16	T	B9
TCATGTTG	IGVFSM1916FXAR,IGVFSM4132TDOU	070_NODJ_10F_03	16	R	B9
ACTGCTCT	IGVFSM1916FXAR,IGVFSM4132TDOU	070_NODJ_10F_03	16	T	B9
ATTTTGCC	IGVFSM2795QIAK,IGVFSM3360ORUF	021_B6J_10M_03	16	R	B10
CTCTAATC	IGVFSM2795QIAK,IGVFSM3360ORUF	021_B6J_10M_03	16	T	B10
ATTTTGCC	IGVFSM2219SJUB,IGVFSM9423CHGE	071_NODJ_10M_03	16	R	B10
CTCTAATC	IGVFSM2219SJUB,IGVFSM9423CHGE	071_NODJ_10M_03	16	T	B10
CTTCTGTA	IGVFSM0587AFGX,IGVFSM7596ZIPA	024_B6J_10F_03	16	R	B11
ACCCTTGC	IGVFSM0587AFGX,IGVFSM7596ZIPA	024_B6J_10F_03	16	T	B11
CTTCTGTA	IGVFSM0749UUSU,IGVFSM1724CQMW	072_NODJ_10F_03	16	R	B11
ACCCTTGC	IGVFSM0749UUSU,IGVFSM1724CQMW	072_NODJ_10F_03	16	T	B11
GTCCATCT	IGVFSM1457MNSR,IGVFSM4108RLII	025_B6J_10M_03	16	R	B12
ATCTTAGG	IGVFSM1457MNSR,IGVFSM4108RLII	025_B6J_10M_03	16	T	B12

GTCCATCT	IGVFSM5506IVPG,IGVFSM9890CMQT	073_NODJ_10M_03	16	R	B12
ATCTTAGG	IGVFSM5506IVPG,IGVFSM9890CMQT	073_NODJ_10M_03	16	T	B12
GCTATCTC	IGVFSM8801AFPS	026_AJ_10F_16	16	R	C1
CATGTCTC	IGVFSM8801AFPS	026_AJ_10F_16	16	T	C1
TAGTTTCC	IGVFSM8141CEOO	027_AJ_10M_16	16	R	C2
TCATTGCA	IGVFSM8141CEOO	027_AJ_10M_16	16	T	C2
TCCATTAT	IGVFSM4118LWUK	028_AJ_10F_16	16	R	C3
ACACCTTT	IGVFSM4118LWUK	028_AJ_10F_16	16	T	C3
AGGATTAA	IGVFSM2182IYXX	031_AJ_10M_16	16	R	C4
AATTTCTC	IGVFSM2182IYXX	031_AJ_10M_16	16	T	C4
AATCTTTC	IGVFSM5510EXRQ	030_AJ_10F_16	16	R	C5
ATTCATGG	IGVFSM5510EXRQ	030_AJ_10F_16	16	T	C5
GTCATATG	IGVFSM4001HPLV	033_AJ_10M_16	16	R	C6
ACTTTACC	IGVFSM4001HPLV	033_AJ_10M_16	16	T	C6
GTGCTTCC	IGVFSM2425SGDB	034_AJ_10F_16	16	R	C7
CTTCTAAC	IGVFSM2425SGDB	034_AJ_10F_16	16	T	C7
ATGTGTTG	IGVFSM7990NFIW	035_AJ_10M_16	16	R	C8
CTATTTCA	IGVFSM7990NFIW	035_AJ_10M_16	16	T	C8
CCATCTTG	IGVFSM2423BNZZ,IGVFSM4370URZQ	026_AJ_10F_03	16	R	C9
TCTCATGC	IGVFSM2423BNZZ,IGVFSM4370URZQ	026_AJ_10F_03	16	T	C9
CCATCTTG	IGVFSM0918MYKM,IGVFSM7550NDQP	076_PWKJ_10F_03	16	R	C9
TCTCATGC	IGVFSM0918MYKM,IGVFSM7550NDQP	076_PWKJ_10F_03	16	T	C9
TACTGTCT	IGVFSM2957XPNR,IGVFSM7221OIET	029_AJ_10M_03	16	R	C10
ATCCTTAC	IGVFSM2957XPNR,IGVFSM7221OIET	029_AJ_10M_03	16	T	C10
TACTGTCT	IGVFSM0563DEOO,IGVFSM6013VRIK	077_PWKJ_10M_03	16	R	C10
ATCCTTAC	IGVFSM0563DEOO,IGVFSM6013VRIK	077_PWKJ_10M_03	16	T	C10
TTCATCGC	IGVFSM7981HLTV,IGVFSM8529NIEC	028_AJ_10F_03	16	R	C11
TAAATATC	IGVFSM7981HLTV,IGVFSM8529NIEC	028_AJ_10F_03	16	T	C11
TTCATCGC	IGVFSM0623LMKH,IGVFSM6356IYPX	078_PWKJ_10F_03	16	R	C11
TAAATATC	IGVFSM0623LMKH,IGVFSM6356IYPX	078_PWKJ_10F_03	16	T	C11
ACTGTGGG	IGVFSM6144DRYM,IGVFSM8120ZQGC	031_AJ_10M_03	16	R	C12
TTACCTGC	IGVFSM6144DRYM,IGVFSM8120ZQGC	031_AJ_10M_03	16	T	C12
ACTGTGGG	IGVFSM2048SMTE,IGVFSM4224SVKL	079_PWKJ_10M_03	16	R	C12
TTACCTGC	IGVFSM2048SMTE,IGVFSM4224SVKL	079_PWKJ_10M_03	16	T	C12
TCTGTGCC	IGVFSM2436WWNM	076_PWKJ_10F_16	16	R	D1

CACTTTCA	IGVFSM2436WWNM	076_PWKJ_10F_16	16	T	D1
TCAATCTC	IGVFSM8807FWBB	077_PWKJ_10M_16	16	R	D2
CACCTTTA	IGVFSM8807FWBB	077_PWKJ_10M_16	16	T	D2
GTCCTCTG	IGVFSM9269MUDI	078_PWKJ_10F_16	16	R	D3
CTGACTTC	IGVFSM9269MUDI	078_PWKJ_10F_16	16	T	D3
TTACATTC	IGVFSM2389EAOO	079_PWKJ_10M_16	16	R	D4
CATTTGGA	IGVFSM2389EAOO	079_PWKJ_10M_16	16	T	D4
ATTCTGTC	IGVFSM0334GXDS	080_PWKJ_10F_16	16	R	D5
GCTCTACT	IGVFSM0334GXDS	080_PWKJ_10F_16	16	T	D5
TGTGTATG	IGVFSM4237VCZG	081_PWKJ_10M_16	16	R	D6
GTTACGTA	IGVFSM4237VCZG	081_PWKJ_10M_16	16	T	D6
TCCATTTG	IGVFSM1365IGJT	084_PWKJ_10F_16	16	R	D7
CCTGTTGC	IGVFSM1365IGJT	084_PWKJ_10F_16	16	T	D7
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GTGCTTGA	IGVFSM1463UXNU,IGVFSM3905UDST	080_PWKJ_10F_03	16	R	D9
GCTATCAT	IGVFSM1463UXNU,IGVFSM3905UDST	080_PWKJ_10F_03	16	T	D9
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TTCGCTAC	IGVFSM3195CJCF,IGVFSM6726YGIG	032_AJ_10F_03	16	T	D11
GAAATTAG	IGVFSM2111IGAF,IGVFSM7111UROK	084_PWKJ_10F_03	16	R	D11
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GCAAATTC	IGVFSM1227KXDL,IGVFSM5036ECAW	085_PWKJ_10M_03	16	R	D12
CATTCTAC	IGVFSM1227KXDL,IGVFSM5036ECAW	085_PWKJ_10M_03	16	T	D12
GAGGTTGA	IGVFSM6869OUVF	042_129S1J_10F_16	16	R	E1
CACTTATC	IGVFSM6869OUVF	042_129S1J_10F_16	16	T	E1
CCTGTCTG	IGVFSM5187OEBL	037_129S1J_10M_16	16	R	E2
ATAAGCTC	IGVFSM5187OEBL	037_129S1J_10M_16	16	T	E2

GTGGGTTC	IGVFSM7768VTDI	038_129S1J_10F_16	16	R	E3
TCATCCTG	IGVFSM7768VTDI	038_129S1J_10F_16	16	T	E3
TTTGCATC	IGVFSM1059YXHB	041_129S1J_10M_16	16	R	E4
CCTGGTAT	IGVFSM1059YXHB	041_129S1J_10M_16	16	T	E4
AGGTAATA	IGVFSM5139SKXC	040_129S1J_10F_16	16	R	E5
TGGTATAC	IGVFSM5139SKXC	040_129S1J_10F_16	16	T	E5
GTGCCTTC	IGVFSM7373DCLC	043_129S1J_10M_16	16	R	E6
TTGGGAGA	IGVFSM7373DCLC	043_129S1J_10M_16	16	T	E6
ATGTTTCC	IGVFSM4259FCLE	044_129S1J_10F_16	16	R	E7
ACTTCATC	IGVFSM4259FCLE	044_129S1J_10F_16	16	T	E7
CTTAATTC	IGVFSM7248MKIA	045_129S1J_10M_16	16	R	E8
TCTCTAGC	IGVFSM7248MKIA	045_129S1J_10M_16	16	T	E8
TCTGGCTC	IGVFSM8124ZHLJ,IGVFSM9227EIZD	036_129S1J_10F_03	16	R	E9
ATGCCCTT	IGVFSM8124ZHLJ,IGVFSM9227EIZD	036_129S1J_10F_03	16	T	E9
TCTGGCTC	IGVFSM7195PKWY,IGVFSM7666QOZY	086_CASTJ_10F_03	16	R	E9
ATGCCCTT	IGVFSM7195PKWY,IGVFSM7666QOZY	086_CASTJ_10F_03	16	T	E9
CATCATTT	IGVFSM0860OBUU,IGVFSM2419NKNV	037_129S1J_10M_03	16	R	E10
CCCAATTT	IGVFSM0860OBUU,IGVFSM2419NKNV	037_129S1J_10M_03	16	T	E10
CATCATTT	IGVFSM7317YYMK,IGVFSM8253LIZU	087_CASTJ_10M_03	16	R	E10
CCCAATTT	IGVFSM7317YYMK,IGVFSM8253LIZU	087_CASTJ_10M_03	16	T	E10
GTTGTCTC	IGVFSM0225XPQG,IGVFSM1783ISPH	038_129S1J_10F_03	16	R	E11
ACTATATA	IGVFSM0225XPQG,IGVFSM1783ISPH	038_129S1J_10F_03	16	T	E11
GTTGTCTC	IGVFSM3708BQMH,IGVFSM8386KIXU	088_CASTJ_10F_03	16	R	E11
ACTATATA	IGVFSM3708BQMH,IGVFSM8386KIXU	088_CASTJ_10F_03	16	T	E11
ATCTTCTG	IGVFSM4535GZXJ,IGVFSM9603NVUP	041_129S1J_10M_03	16	R	E12
CTCTATAC	IGVFSM4535GZXJ,IGVFSM9603NVUP	041_129S1J_10M_03	16	T	E12
ATCTTCTG	IGVFSM6275QJJU,IGVFSM7470VZNE	089_CASTJ_10M_03	16	R	E12
CTCTATAC	IGVFSM6275QJJU,IGVFSM7470VZNE	089_CASTJ_10M_03	16	T	E12
TGTTTGCC	IGVFSM0725SKMI	086_CASTJ_10F_16	16	R	F1
CTGTCTCA	IGVFSM0725SKMI	086_CASTJ_10F_16	16	T	F1
TTCTGTCA	IGVFSM8092ISYK	087_CASTJ_10M_16	16	R	F2
GACCTTTC	IGVFSM8092ISYK	087_CASTJ_10M_16	16	T	F2
ACGGACTC	IGVFSM3158KFTN	092_CASTJ_10F_16	16	R	F3
GATTTGGC	IGVFSM3158KFTN	092_CASTJ_10F_16	16	T	F3
TTTGGTCA	IGVFSM2176DUOO	089_CASTJ_10M_16	16	R	F4

CGTCTAGG	IGVFSM2176DUOO	089_CASTJ_10M_16	16	T	F4
TATCCGGG	IGVFSM7844SUGB	090_CASTJ_10F_16	16	R	F5
TACTCGAA	IGVFSM7844SUGB	090_CASTJ_10F_16	16	T	F5
TGTCATTC	IGVFSM1128BAML	091_CASTJ_10M_16	16	R	F6
CAGCCTTT	IGVFSM1128BAML	091_CASTJ_10M_16	16	T	F6
ATTCTCTG	IGVFSM4593QSDG	094_CASTJ_10F_16	16	R	F7
CCTCATTA	IGVFSM4593QSDG	094_CASTJ_10F_16	16	T	F7
TGGCTTCC	IGVFSM3110AEWG	093_CASTJ_10M_16	16	R	F8
CTTATACC	IGVFSM3110AEWG	093_CASTJ_10M_16	16	T	F8
TTGTTGCC	IGVFSM1865RBAP,IGVFSM3592ZCYS	040_129S1J_10F_03	16	R	F9
TCTATTAC	IGVFSM1865RBAP,IGVFSM3592ZCYS	040_129S1J_10F_03	16	T	F9
TTGTTGCC	IGVFSM3391TANO,IGVFSM9194WULA	090_CASTJ_10F_03	16	R	F9
TCTATTAC	IGVFSM3391TANO,IGVFSM9194WULA	090_CASTJ_10F_03	16	T	F9
GTCATCTC	IGVFSM2592YCXF,IGVFSM4046THJJ	043_129S1J_10M_03	16	R	F10
CCTGCATT	IGVFSM2592YCXF,IGVFSM4046THJJ	043_129S1J_10M_03	16	T	F10
GTCATCTC	IGVFSM3523HTHO,IGVFSM4618XBBU	091_CASTJ_10M_03	16	R	F10
CCTGCATT	IGVFSM3523HTHO,IGVFSM4618XBBU	091_CASTJ_10M_03	16	T	F10
TTGCTCAT	IGVFSM0923PDGB,IGVFSM8263PWJC	044_129S1J_10F_03	16	R	F11
CAATCCTT	IGVFSM0923PDGB,IGVFSM8263PWJC	044_129S1J_10F_03	16	T	F11
TTGCTCAT	IGVFSM7747QNMD,IGVFSM8043HCTC	094_CASTJ_10F_03	16	R	F11
CAATCCTT	IGVFSM7747QNMD,IGVFSM8043HCTC	094_CASTJ_10F_03	16	T	F11
CTGTCTGC	IGVFSM0221ZKMO,IGVFSM9934DIHO	045_129S1J_10M_03	16	R	F12
TTGTCTTA	IGVFSM0221ZKMO,IGVFSM9934DIHO	045_129S1J_10M_03	16	T	F12
CTGTCTGC	IGVFSM0611PVMT,IGVFSM8601VZGE	093_CASTJ_10M_03	16	R	F12
TTGTCTTA	IGVFSM0611PVMT,IGVFSM8601VZGE	093_CASTJ_10M_03	16	T	F12
TATATTCC	IGVFSM0217YUAM	058_WSBJ_10F_16	16	R	G1
TCACTTTA	IGVFSM0217YUAM	058_WSBJ_10F_16	16	T	G1
ATATTGGC	IGVFSM5571BGFV	057_WSBJ_10M_16	16	R	G2
TGCTTGGG	IGVFSM5571BGFV	057_WSBJ_10M_16	16	T	G2
GTGTCCTC	IGVFSM0943LRWD	064_WSBJ_10F_16	16	R	G3
CGCTCATT	IGVFSM0943LRWD	064_WSBJ_10F_16	16	T	G3
ATCTTCAT	IGVFSM8559RKMK	059_WSBJ_10M_16	16	R	G4
GCCTCTAT	IGVFSM8559RKMK	059_WSBJ_10M_16	16	T	G4
CGTGGTTG	IGVFSM2242OSDW	062_WSBJ_10F_16	16	R	G5
GAGCACAA	IGVFSM2242OSDW	062_WSBJ_10F_16	16	T	G5

TTGCATCC	IGVFSM7709RATJ	061_WSBJ_10M_16	16	R	G6
CTCTTAAC	IGVFSM7709RATJ	061_WSBJ_10M_16	16	T	G6
TCTTAATC	IGVFSM4280ZZUH	096_WSBJ_10F_16	16	R	G7
TCTAGGCT	IGVFSM4280ZZUH	096_WSBJ_10F_16	16	T	G7
TGCATTTC	IGVFSM5083IADN	063_WSBJ_10M_16	16	R	G8
AATTCTGC	IGVFSM5083IADN	063_WSBJ_10M_16	16	T	G8
GATGTTTC	IGVFSM1395QLCT,IGVFSM2105LIWI	058_WSBJ_10F_03	16	R	G9
CATTCTCA	IGVFSM1395QLCT,IGVFSM2105LIWI	058_WSBJ_10F_03	16	T	G9
GATGTTTC	IGVFSM0322EOHO,IGVFSM4189AKUL	092_CASTJ_10F_03	16	R	G9
CATTCTCA	IGVFSM0322EOHO,IGVFSM4189AKUL	092_CASTJ_10F_03	16	T	G9
ATCTTGTC	IGVFSM7371QPPK,IGVFSM9072COND	047_NZOJ_10M_03	16	R	G10
ACTTGCCT	IGVFSM7371QPPK,IGVFSM9072COND	047_NZOJ_10M_03	16	T	G10
ATCTTGTC	IGVFSM1203GHYB,IGVFSM6575IFHC	057_WSBJ_10M_03	16	R	G10
ACTTGCCT	IGVFSM1203GHYB,IGVFSM6575IFHC	057_WSBJ_10M_03	16	T	G10
TCATATTC	IGVFSM8490EBUM,IGVFSM9852CDAM	048_NZOJ_10F_03	16	R	G11
ATCATTGC	IGVFSM8490EBUM,IGVFSM9852CDAM	048_NZOJ_10F_03	16	T	G11
TCATATTC	IGVFSM1996USDH,IGVFSM9916UWTV	060_WSBJ_10F_03	16	R	G11
ATCATTGC	IGVFSM1996USDH,IGVFSM9916UWTV	060_WSBJ_10F_03	16	T	G11
TGGCCTCT	IGVFSM0101ITFE,IGVFSM4528ONTR	049_NZOJ_10M_03	16	R	G12
GTTCAACA	IGVFSM0101ITFE,IGVFSM4528ONTR	049_NZOJ_10M_03	16	T	G12
TGGCCTCT	IGVFSM1116FVJP,IGVFSM1295CUJX	059_WSBJ_10M_03	16	R	G12
GTTCAACA	IGVFSM1116FVJP,IGVFSM1295CUJX	059_WSBJ_10M_03	16	T	G12
CGTTGTCT	IGVFSM5504RVNQ	046_NZOJ_10F_16	16	R	H1
CCATTTGC	IGVFSM5504RVNQ	046_NZOJ_10F_16	16	T	H1
TCTTGTCA	IGVFSM1584RGVM	047_NZOJ_10M_16	16	R	H2
GACTTTGC	IGVFSM1584RGVM	047_NZOJ_10M_16	16	T	H2
TATTCCTG	IGVFSM8444KPOJ	048_NZOJ_10F_16	16	R	H3
ATTGGCTC	IGVFSM8444KPOJ	048_NZOJ_10F_16	16	T	H3
TCCATGTC	IGVFSM4489MUQD	049_NZOJ_10M_16	16	R	H4
GTGCTAGC	IGVFSM4489MUQD	049_NZOJ_10M_16	16	T	H4
TTGTCATC	IGVFSM5314ZJEY	050_NZOJ_10F_16	16	R	H5
CTTTCAAC	IGVFSM5314ZJEY	050_NZOJ_10F_16	16	T	H5
ATTTCCTG	IGVFSM1469JLYZ	051_NZOJ_10M_16	16	R	H6
ACTATTGC	IGVFSM1469JLYZ	051_NZOJ_10M_16	16	T	H6
GTGTCTCC	IGVFSM0037THNT	052_NZOJ_10F_16	16	R	H7

ACTGGCTT	IGVFSM0037THNT	052_NZOJ_10F_16	16	T	H7
GTGTGTGT	IGVFSM5658GGTC	053_NZOJ_10M_16	16	R	H8
ATTAGGCT	IGVFSM5658GGTC	053_NZOJ_10M_16	16	T	H8
TATGCTTC	IGVFSM0561GQRI,IGVFSM5306QAYC	050_NZOJ_10F_03	16	R	H9
GCCTTTCA	IGVFSM0561GQRI,IGVFSM5306QAYC	050_NZOJ_10F_03	16	T	H9
TATGCTTC	IGVFSM2903TFAJ,IGVFSM5785PIAM	062_WSBJ_10F_03	16	R	H9
GCCTTTCA	IGVFSM2903TFAJ,IGVFSM5785PIAM	062_WSBJ_10F_03	16	T	H9
ATGGTGTT	IGVFSM1083MRIY,IGVFSM5958RVMJ	051_NZOJ_10M_03	16	R	H10
ATTCTAGG	IGVFSM1083MRIY,IGVFSM5958RVMJ	051_NZOJ_10M_03	16	T	H10
ATGGTGTT	IGVFSM0431DSEH,IGVFSM0898KHQX	061_WSBJ_10M_03	16	R	H10
ATTCTAGG	IGVFSM0431DSEH,IGVFSM0898KHQX	061_WSBJ_10M_03	16	T	H10
GAATAATG	IGVFSM6096DCNO,IGVFSM8959FRNP	052_NZOJ_10F_03	16	R	H11
CCTTACAT	IGVFSM6096DCNO,IGVFSM8959FRNP	052_NZOJ_10F_03	16	T	H11
GAATAATG	IGVFSM1901NWCD,IGVFSM7994PCAR	096_WSBJ_10F_03	16	R	H11
CCTTACAT	IGVFSM1901NWCD,IGVFSM7994PCAR	096_WSBJ_10F_03	16	T	H11
CCTCTGTG	IGVFSM8856ZHGA,IGVFSM9234QGBN	053_NZOJ_10M_03	16	R	H12
ACATTTGG	IGVFSM8856ZHGA,IGVFSM9234QGBN	053_NZOJ_10M_03	16	T	H12
CCTCTGTG	IGVFSM1039KCGC,IGVFSM8727ZEXW	063_WSBJ_10M_03	16	R	H12
ACATTTGG	IGVFSM1039KCGC,IGVFSM8727ZEXW	063_WSBJ_10M_03	16	T	H12

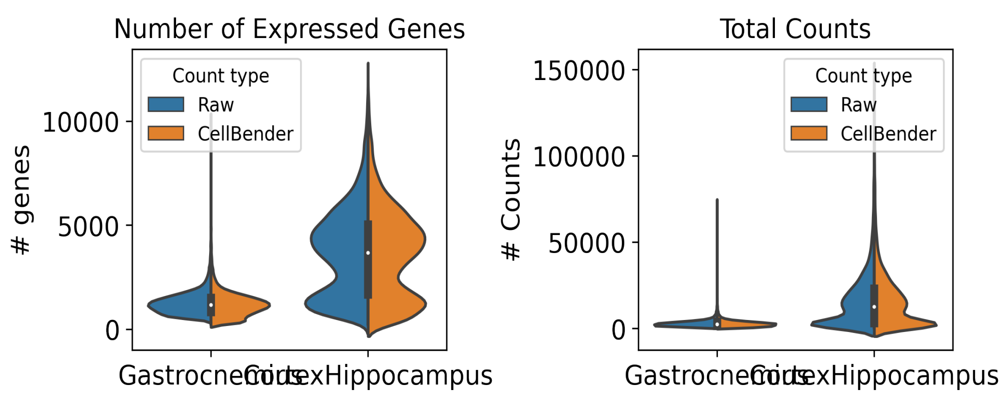
6. Description of data processing workflow and h5ad files

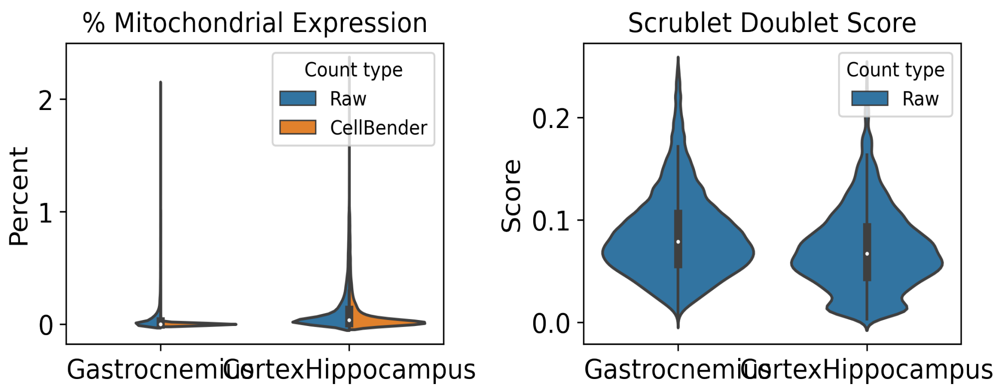
Data from Subpool_13 were combined with all other subpools into one AnnData object per tissue as well as matching data from other plates for processing and cell type annotation. The AnnData contains detailed sample and cell metadata in the observation (obs) table and gene information in the variables table (var). Cell IDs (adata.obs.index) are re-formatted to be human-readable and unique across subpools and experiments by appending subpool and experiment IDs. The final AnnData contains both raw and CellBender denoised counts matrices in separate layers: adata.layers['raw_counts'] and adata.layers['cellbender_counts']. The current adata.X points to the CellBender denoised counts. Cells or nuclei are filtered by >0.5 cell_probability from CellBender analysis.

In addition to CellBender filtering, nuclei were also filtered by the following QC parameters:

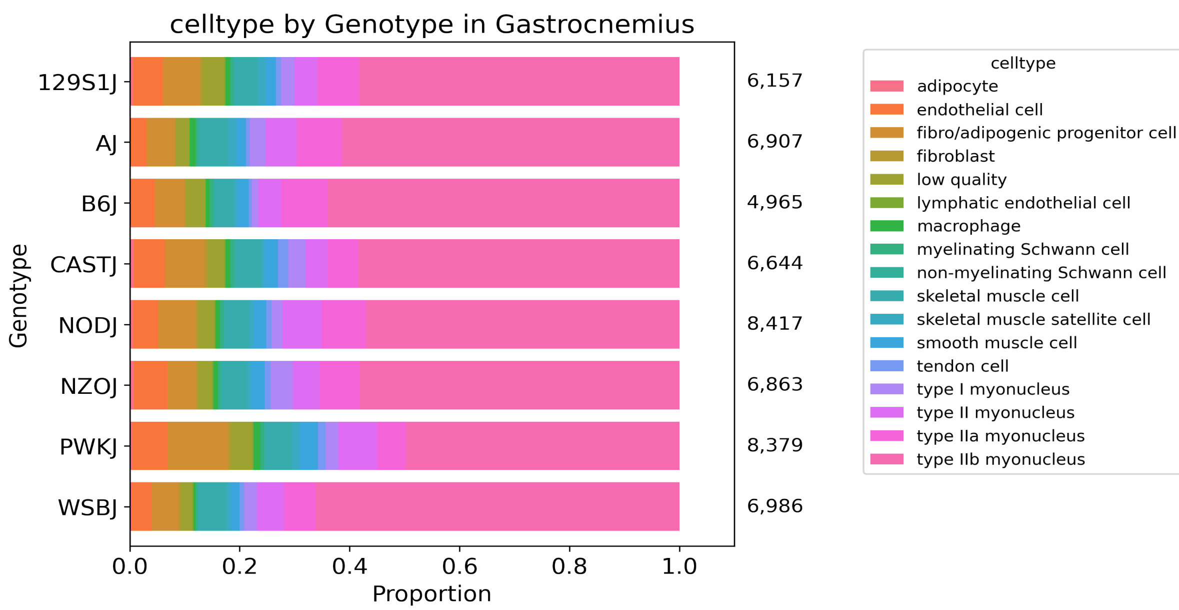
Table 8: QC thresholds

Parameter	Threshold
Minimum # UMIs	500.0
Maximum # UMIs	150000.0
Minimum # genes	250.0
Maximum % mito.	1.0
Maximum doublet score	0.25





After filtering, data for each tissue was normalized and clustered using CellBender denoised counts. Normalization includes regression of technical parameters (number of genes expressed per cell and percent mitochondrial gene expression). Harmony was used to integrate exome capture and non-exome capture subpools for annotation. A Leiden clustering resolution of 1 was used for 36 total clusters in Gastrocnemius and 55 clusters in CortexHippocampus, using all tissue data across experiments. Cell type annotations were assigned per Leiden cluster or subcluster (leiden_R) using expression of canonical cell type marker genes.



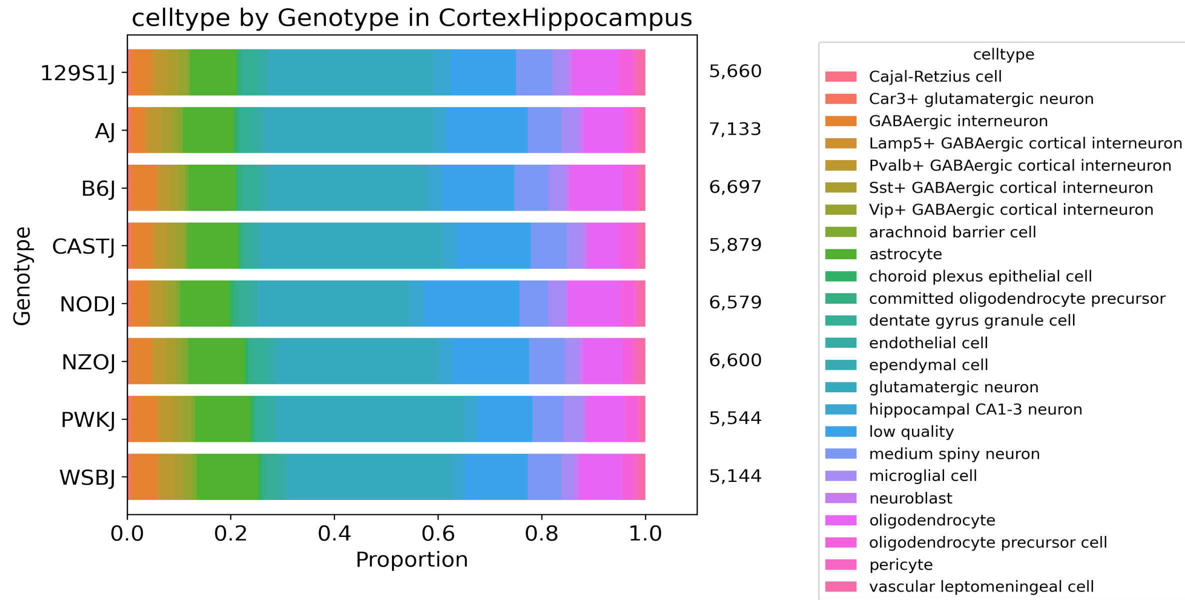


Table 9: Description of obs keys in h5ad file

Key	Description
cellID	Index of obs table. Includes barcoding wells for each round (1_2_3), subpool, and experiment.
lab_sample_id	Human-readable sample name used in-house.
sample	IGVF tissue accession.
plate	Experiment ID, e.g. igvf_003.
subpool	Subpool ID, e.g. Subpool_2. Number corresponds to Illumina index ID from Parse Bio kit.
SampleType	Nuclei or Cells.
Tissue	Human-readable tissue name, e.g. Kidney.
Sex	Male or Female.
Age	Age in postnatal months (PNM) of the mouse, e.g. PNM_02.
Genotype	Mouse genotype/strain, e.g. B6J.
leiden	Numeric Leiden cluster starting from 0.
leiden_R	If necessary, Leiden clusters and any sub-clusters, e.g. 15_1.
subpool_type	Exome capture (EX) or no exome capture (NO).
general_celltype	Broadest level of cell type annotations, e.g. neuron.
general_CL_ID	EMBL CL database ID for general cell type annotation, e.g. CL:0000540.
celltype	Finest level of cell type annotation with a CL ID, e.g. Vip+ GABAergic cortical interneuron.
CL_ID	EMBL CL database ID for cell type annotation, CL:4023016
subtype	Finest level of cell type annotation that may not have a CL ID. Mostly matches celltype.
Protocol	Cell barcoding and library building protocol/kit, e.g. Parse_WT.

Chemistry	Protocol chemistry version, v2 or v3.
bc	24-nucleotide sequence corresponding to 3 8-nucleotide barcode rounds (3, 2, 1).
bc1_sequence	8-nucleotide sequence barcode for round 1.
bc2_sequence	8-nucleotide sequence barcode for round 2.
bc3_sequence	8-nucleotide sequence barcode for round 3.
bc1_well	Round 1 well.
bc2_well	Round 2 well.
bc3_well	Round 2 well.
Row	Sample barcoding plate (round 1) row.
Column	Sample barcoding plate (round 1) column.
well_type	Single or Multiplexed.
Mouse_Tissue_ID	Same as lab_sample_id.
Multiplexed_sample1	Sample 1 from multiplexed well (if applicable)
Multiplexed_sample2	Sample 2 from multiplexed well (if applicable)
DOB	Date of birth of the mouse in month/day/year format.
Age_days	Age of the mouse in days.
Body_weight_g	Body weight in grams of the mouse.
Estrus_cycle	Estimated estrus stage of the mouse, if applicable.
Dissection_date	Date the tissue was harvested in month/day/year format.
Dissection_time	Time in hours:minutes (24-hour time) the mouse was sacrificed.
ZT	Number of hours into the light cycle, from lights on (06:30, ZT0) to lights off (20:30, ZT14).
Dissector	Initials of technician who dissected the tissue.
Tissue_weight_mg	Tissue weight in milligrams.
Notes	Notes taken during sample collection, experiment, or data processing.
n_genes_by_counts_raw	The number of genes with at least 1 count, calculated using raw counts.
total_counts_raw	Total counts (UMIs), calculated using raw counts.
total_counts_mt_raw	Total counts for mitochondrial genes, calculated using raw counts.
pct_counts_mt_raw	Percent of total counts in mitochondrial genes, calculated using raw counts.
n_genes_by_counts_cb	The number of genes with at least 1 count, calculated using CellBender counts.
total_counts_cb	Total counts (UMIs), calculated using CellBender counts.
total_counts_mt_cb	Total counts for mitochondrial genes, calculated using CellBender counts.
pct_counts_mt_cb	Percent of total counts in mitochondrial genes, calculated using CellBender counts.
doublet_score	Scrublet doublet score, calculated using CellBender counts.
predicted_doublet	Scrublet predicted doublet (True or False).
background_fraction	Fraction of counts CellBender predicted to be background noise.

cell_probability	CellBender probability that the cell is real.
cell_size	CellBender size factor.
droplet_efficiency	CellBender statistic indicating transcript capture efficiency per cell.
B6J_klue_counts	Estimated counts for B6J genotype from klue analysis.
NODJ_klue_counts	Estimated counts for NODJ genotype from klue analysis.
AJ_klue_counts	Estimated counts for AJ genotype from klue analysis.
PWKJ_klue_counts	Estimated counts for PWKJ genotype from klue analysis.
129S1J_klue_counts	Estimated counts for 129S1J genotype from klue analysis.
CASTJ_klue_counts	Estimated counts for CASTJ genotype from klue analysis.
WSBJ_klue_counts	Estimated counts for WSBJ genotype from klue analysis.
NZOJ_klue_counts	Estimated counts for NZOJ genotype from klue analysis.

7. Package versions

Table 10: Package versions

package	version
kallisto	0.50.1
bustools	0.43.2
kb	0.28.2
Python	3.9.0
snakemake	7.32.0
cellbender	0.3.2
scanpy	1.10.2
scrublet	0.2.3
anndata	0.10.8
pandas	2.2.2
numpy	1.26.4
harmonypy	0.0.9

8. Contact information and GitHub

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Pipeline repo: https://github.com/mortazavilab/parse_pipeline

Code used to make this report: https://github.com/mortazavilab/8cube_manuscript/blob/main/plate_report/Subpool_report_igvf_011.ipynb

Code used to process and annotate the data for main tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Gastrocnemius.ipynb

Code used to process and annotate the data for multiplexed tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/CortexHippocampus.ipynb